LaTeX Feature - One-Pager

LaTeX & Research Tools with Live Reloading

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Branch: `feature/latex-research-tools-live-reload`

Status: ✅ Core Implementation Complete

What We Built

1. Live Reloading Development Server

File: `scripts/devresearchserver.py`

A development server that automatically restarts when code changes, enabling rapid iterative development of LaTeX and research tools.

Features:

- Automatic server restart on file changes
- Watches `researchsimulationserver.py` and `src/waft/evolution/`
- Auto-opens browser to http://localhost:8001
- Development mode indicator

Usage:

```
python3 scripts/dev_research_server.py
```

2. LaTeX Generator Module

File: `src/waft/evolution/latex_generator.py`

A complete LaTeX generation system integrated with WAFT's evolution framework.

Integration Points:

- **ChatDistiller**: Extracts structured ideas from content
- **StylingGenome**: Applies WAFT styling presets (clinical_standard, premium, professional)
- **StylingGenomeRegistry**: Access to style configurations

Features:

- Markdown to LaTeX conversion
- LaTeX character escaping
- Multiple document classes (article, report, book)
- Custom packages and preamble support
- Optional PDF compilation via pdflatex

Usage:

```
from src.waft.evolution.latex_generator import generate_latex

generate_latex(
    content="# My Document\n\nContent here...",
    title="My Document",
    document_class="article",
    style="clinical_standard"
)
```

3. Research Server Integration

New Endpoints:

- `GET /api/report/latex` - Download research report as LaTeX
- `POST /api/export/latex` - Export any content as LaTeX

UI Updates:

- Added "📝 Export as LaTeX" button alongside PDF report link
- Available when simulation is complete

Why This Matters

Using WAFT to Develop WAFT

This feature demonstrates **"eating our own dog food"** - using WAFT's tools to develop WAFT features:

1. **Live Reloading**: Enables rapid iteration on LaTeX capabilities
2. **LaTeX Generation**: Adds new output format to WAFT's ecosystem
3. **Research Tools**: Enhances documentation and analysis capabilities
4. **Integration**: Works seamlessly with existing WAFT systems

Technical Architecture

```
LaTeXGenerator
├── ChatDistiller Integration
│   └── Extracts structured ideas (concepts, decisions, insights, actions)
├── StylingGenome Integration
│   └── Applies WAFT styling presets (fonts, margins, colors)
├── Markdown to LaTeX Conversion
│   └── Headers, lists, code blocks, paragraphs
└── PDF Compilation (Optional)
    └── Uses pdflatex for PDF generation
```

Next Steps

1. **Enhanced LaTeX Features**
2. Table generation from data
3. Figure/image support

4. Bibliography support
5. Custom LaTeX templates
6. Math equation support
7. **Research Tools Enhancement**
8. LaTeX export for all research reports
9. Comparative LaTeX generation
10. LaTeX template customization
11. Batch LaTeX export
12. **Development Experience**
13. Hot module reloading (no server restart)
14. WebSocket for live updates
15. Development dashboard
16. Error overlay in browser

Files Created

- ✅ `scripts/devresearchserver.py` - Live reloading server
- ✅ `scripts/researchsimulationserver.py` - Added dev mode and LaTeX endpoints
- ✅ `src/waft/evolution/latex_generator.py` - LaTeX generator (500+ lines)
- ✅ `src/waft/evolution/init.py` - Added LaTeX exports
- ✅ `workefforts/CHECKPOINT2026-01-11LATEXRESEARCHTOOLSLIVERELOAD.md` - Documentation

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